Primary Dysmenorrhea

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Abstract. Managing primary dysmenorrhea (PD) requires understanding its complex mechanisms involving prostaglandins, oxidative stress, and inflammation.

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Overview

Primary dysmenorrhea (PD) is defined as painful menstruation occurring in the absence of any underlying pelvic disease or pathology. It is the most common gynecological complaint affecting women of reproductive age, with a prevalence ranging from 45% to 95% worldwide.[1][2] Typically, the pain starts just before or at the onset of menstruation and lasts for up to 72 hours. It is characterized by cramping pain in the lower abdomen, which is often accompanied by systemic symptoms such as nausea, vomiting, fatigue, headaches, and gastrointestinal distress.[3] The underlying pathophysiology of PD is primarily attributed to the overproduction of prostaglandins (specifically PGF2α and PGE2) in the endometrial tissue.[4] This leads to uterine hypercontractility, ischemia, and hypoxia, which, in turn, cause the painful cramps experienced during menstruation.[5] Despite its high prevalence, PD remains underdiagnosed and undertreated, often being normalized as a “normal” part of menstruation.[6] The microbiome plays a significant, yet underexplored, role in the inflammatory processes associated with PD. Studies have suggested that dysbiosis or imbalance in the vaginal microbiome could exacerbate inflammation, although further research is required to establish these connections fully.

Associated Conditions

Primary dysmenorrhea is closely linked to other conditions, both gynecological and non-gynecological. The most significant associated condition is endometriosis, where endometrial-like tissue grows outside the uterus, causing severe pain similar to PD.[7][8] Adolescents and young women with PD are at an increased risk of developing endometriosis in adulthood, especially when their dysmenorrhea is severe and persistent.[9][10] Psychologically, depression and anxiety are prevalent among women with PD.[11][12] A study reported that women with PD have a significantly higher risk of developing depression, with up to 64% of women with PD being at risk of postpartum depression (PPD) following childbirth.[13] The chronic pain experienced in PD leads to a decreased quality of life, contributing to higher rates of absenteeism from work and school, further exacerbating mental health challenges.[14] Other conditions that can overlap with or mimic PD include irritable bowel syndrome (IBS) and pelvic floor dysfunction, which can complicate the diagnosis and management of PD.

Causal Theories

The etiology of primary dysmenorrhea is largely attributed to increased levels of prostaglandins in the menstrual fluid. These substances promote uterine contractions and vasoconstriction, leading to ischemia and pain. However, the full range of contributing factors is still not completely understood. Some of the key theories include:

Causal Theory	Explanation
Prostaglandin Overproduction	Prostaglandins cause uterine contractions, leading to pain and ischemia.[15][x]
Microbiome Dysbiosis	The microbiome may affect inflammatory responses during menstruation, potentially exacerbating PD symptoms.[16][17]
Genetic Factors	Genetic predispositions may influence the severity and onset of dysmenorrhea.[18]

Diagnosis

The diagnosis of primary dysmenorrhea is primarily clinical, based on the patient's medical history and symptom presentation. It is essential to differentiate PD from secondary dysmenorrhea, which can be caused by underlying pelvic conditions such as endometriosis and adenomyosis.[19] Secondary causes of dysmenorrhea often present later in life and are not cyclical. Advancements in non-invasive diagnostic methods, including the use of microbiome signatures, metabolomics, and metallomics, have shown potential in identifying biomarkers that may assist in diagnosing dysmenorrhea. However, these methods are not yet widely used in clinical settings.

Primer

To fully understand primary dysmenorrhea, it is crucial to grasp the fundamentals of its pathophysiology, which is largely driven by the release of prostaglandins.[20][x] These molecules mediate uterine contractions and vasoconstriction during menstruation, leading to pain. The elevated levels of prostaglandins, especially PGF2α and PGE2, cause intense uterine muscle contractions that reduce blood flow to the uterus, leading to ischemia, hypoxia, and a heightened sensitivity to pain.[21] This process is further exacerbated by inflammatory responses and oxidative stress in the uterine tissues. As we deepen our understanding of PD, it is important to explore not only the hormonal and inflammatory factors but also the roles of heavy metals and metabolites in the condition's pathogenesis. Heavy metals, which accumulate in the body through environmental exposure, can disrupt normal physiological processes, including inflammatory responses and oxidative stress pathways. For instance, metals like zinc, copper, and iron are essential in numerous biochemical reactions, but imbalances in these metals can exacerbate symptoms of a lot of conditions. Similarly, metabolites can reflect the biochemical alterations occurring during menstruation and the luteal phase, potentially offering insights into the underlying mechanisms driving PD.

Metallomic Signatures

Metal imbalances could serve as useful biomarkers for diagnosing and monitoring PD. The role of heavy metals in primary dysmenorrhea is an emerging area of interest, especially in understanding how metal imbalances contribute to inflammation, oxidative stress, and the pain mechanisms that drive this condition. Recent studies have suggested that essential trace metals such as zinc, copper, and iron play pivotal roles in various physiological processes, and their imbalances may significantly affect uterine function, potentially exacerbating PD symptoms. Zinc, a vital trace element, is one of the most studied metals in the context of PD. It has been demonstrated to exhibit potent anti-inflammatory and antioxidant properties, which are highly important in alleviating the symptoms of PD. Zinc functions through several mechanisms: it inhibits the production of prostaglandins, which are known to cause uterine contractions and pain, by modulating the activity of cyclooxygenases (COX).[22] Zinc helps to enhance microcirculation within the endometrial tissues, preventing ischemia and reducing the severity of uterine cramps. The anti-inflammatory effect of zinc is partly due to its ability to increase the levels of superoxide dismutase (SOD1), an important antioxidant enzyme that reduces the production of free radicals.[23] These actions help mitigate oxidative stress, which is often elevated in PD and contributes to pain and tissue damage.

Metabolomic Signatures

The luteal phase of the menstrual cycle, especially the luteal regression stage, plays a significant role in the development of PD. This stage, marked by a decline in progesterone and the shedding of the endometrial lining, has been shown to produce metabolic and hormonal changes that contribute to the pain and discomfort associated with PD. The use of urine metabolomics has led to the identification of several biomarkers that could serve as indicators of PD during the luteal regression phase. The study on metabolomic profiles of PD patients during this stage revealed significant perturbations in various metabolites. Urinary metabolites such as ornithine, dihydrocor-tisol, histidine, citrulline, sphinganine, and phytosphingosine were identified as being notably altered in women with PD compared to healthy controls.[24] These metabolites are involved in pathways related to inflammation, oxidative stress, and steroid hormone synthesis, processes that are central to PD pathophysiology. Sphinganine and phytosphingosine are sphingolipids involved in cell signaling and inflammation, while citrulline and ornithine are amino acids associated with the urea cycle and nitric oxide synthesis, which could influence vascular tone and uterine contractions. 15-keto-prostaglandin F2α, a metabolite of prostaglandin F2α (PGF2α), is another key biomarker for PD. PGF2α, a potent mediator of uterine contractions and vasoconstriction, is central to the pain and ischemia experienced during menstruation.[25] The elevated levels of 15-keto-prostaglandin F2α, along with the other metabolites, suggest a heightened inflammatory and oxidative state in PD patients, during the luteal regression phase.

Interventions

Primary dysmenorrhea is a common condition that can severely impact quality of life. While traditional pharmaceutical approaches, such as NSAIDs and hormonal treatments, are often used to alleviate symptoms, a range of alternative and complementary therapies have shown potential in managing PD, particularly microbiome-targeted interventions (MBTIs) and systemic inflammation. The interventions discussed here include vitamins, probiotics, omega-3 fatty acids, TENS, metformin, lactoferrin, acupuncture, acupressure, exercise, yoga, massage, calcium, and nifedipine. Each of these therapies has a distinct mechanism of action, with varying degrees of evidence supporting their efficacy in treating PD.

Intervention	Mechanism of Action	MBTI Status
Vitamins (Vitamin D, E)	Vitamin D regulates inflammation by enhancing calcium absorption and modulating (by decreasing) prostaglandin production, reducing uterine cramps.[26] Vitamin E acts as a powerful antioxidant, reducing oxidative stress and inflammation.[27]	Promising Candidate
Probiotics	Probiotics, such as Lactobacillus and Bifidobacterium strains, modulate the gut microbiota, which in turn influences systemic inflammation and immune responses.[28] By regulating inflammatory mediators like cytokines and prostaglandins, probiotics may reduce the severity of PD symptoms.[29]	Under Investigation
Omega-3 Fatty Acids	Omega-3s, particularly EPA and DHA, reduce inflammation by modulating the production of prostaglandins (PGE3), which have anti-inflammatory effects.[30] Omega-3 supplementation reduces the levels of pro-inflammatory cytokines and enhances overall immune response, thus alleviating menstrual pain.	Validated
Metformin	Metformin improves insulin sensitivity and reduces inflammation, which may indirectly decrease prostaglandin production, a key contributor to PD. Studies have shown a drastic decrease in dysmenorrhea after treatment with metformin [31]	Proposed candidate
Lactoferrin	Lactoferrin, a glycoprotein found in milk, has anti-inflammatory properties and may help modulate immune responses.[32] It is thought to reduce menstrual distress by lowering systemic inflammation, potentially influencing microbiome-related pathways to alleviate symptoms.[33]	Under Investigation
Calcium	Calcium plays a crucial role in muscle contraction and relaxation, including the uterine muscles.[34] Supplementation may help alleviate menstrual cramps by improving muscle function and reducing uterine hyperactivity.[35]	Promising Candidate
Nifedipine (drug repurposing)	Nifedipine, a calcium channel blocker, works by relaxing smooth muscle tissue, including the uterine muscles, thereby reducing uterine contractions and alleviating menstrual pain. It may be particularly beneficial for women who experience severe cramps.[36]	Under Investigation

▶ **Other Complementary and Alternative Therapies**

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